Edugenie- Project Development Phases

4. Project Planning Phase

The project planning phase determines how the project will be implemented and divided into manageable activities.

Project Objectives

The main objectives were:

- Develop an AI-powered educational assistant.
- Integrate Generative AI into an educational application.
- Create a simple web interface.
- Develop multiple independent learning modules.
- Connect the frontend with FastAPI APIs.
- Test each feature independently.
- Demonstrate the complete working system.

Development Plan

Phase	Activity
Phase 1	Brainstorming and idea selection
Phase 2	Requirement analysis
Phase 3	System and UI design
Phase 4	Backend development
Phase 5	AI module integration
Phase 6	Frontend development
Phase 7	Testing and debugging
Phase 8	Documentation and demonstration

Technology Planning

The project uses Python-based backend technologies and web technologies for the frontend. Cloud-based AI provides the main generative capabilities while lightweight local processing can be used for selected explanation tasks.

Risk Planning

Possible challenges identified during planning included:

- AI-generated incorrect responses.
- API failures.
- Invalid JSON responses from quiz generation.
- Internet dependency.
- API key security.
- Slow response for some AI operations.
- Compatibility issues between frontend and backend.

Appropriate error handling and testing were planned to reduce these problems.

- **Main.py**

main.py

```python
from fastapi import FastAPI,Request
from fastapi.responses import HTMLResponse
from pydantic import BaseModel
import requests

app = FastAPI()
class Q(BaseModel):
    question: str = ""; text: str = ""; topic: str = ""; subject: str =

def get_answer(q):
    q = q.strip().replace("?","")
    if not q: return "Type a question"
    try:
        r = requests.get(f"https://en.wikipedia.org/api/rest_v1/page/sum
        if "extract" in r: return r["extract"]
        # search pannitu answer edukkum
        s = requests.get("https://en.wikipedia.org/w/api.php", params={'
        title = s["query"]["search"][0]["title"]
```

main.py

```
def get_answer(q):
    try:
    except:
        return f"Real info about '{q}' will appear here. (Check internet

HTML = """<html><body style="font-family:sans-serif;max-width:700px;marg
<h2>Welcome to EduGenie ❤️🧚</h2><p>Your personal AI tutor!</p><hr>
<div style="background:#f0f9ff;padding:15px;margin:10px;border-radius:10
<div style="background:#f0fdf4;padding:15px;margin:10px;border-radius:10
<div style="background:#fefce8;padding:15px;margin:10px;border-radius:10
<div style="background:#fdf2f8;padding:15px;margin:10px;border-radius:10
<div style="background:#f5f3ff;padding:15px;margin:10px;border-radius:10
<script>
async function ask(){let v=document.getElementById('q1').value;S let r=a
async function exp(){let v=document.getElementById('q2').value; let r=aw
async function sum(){let v=document.getElementById('q3').value; let r=aw
async function quiz(){let v=document.getElementById('q4').value; let r=a
async function rec(){let v=document.getElementById('q5').value; let r=aw
</script></body></html>"""
```